



Linux Network Stack Administration

A Developer's Approach

This article explores the various methods of managing the networking stack using application-level C programs, kernel-level C programs, the `proc` filesystem, etc.

Flat, Linux has been gaining popularity in IT-enabled industries. Linux networking has grown side by side with the operating system, and it's very common to see Linux-enabled devices in the networking domain.

Linux provides a very versatile, efficient and well-organised network stack. The TCP/IP implementation of Linux could be considered one of the best amongst operating systems. While packet processing through TCP/IP is a very interesting topic, this article will cover monitoring and managing the Linux TCP/IP stack. Although there are many tools that can be used for the same purpose, this article aims to explore administration of the networking stack without using external tools. This is more useful, especially for developers while working on embedded or remote systems where such 'external' tools are not available.

This article attempts to cover the various methods of managing the networking stack, i.e., by application-level C programs, kernel-level C

programs, the `proc` filesystem, etc.

`proc` is a virtual filesystem that can be located under the `/proc` mount point. This is really not a filesystem, but a window to look into the Linux kernel. The files inside `/proc` are not really files but can be termed as calls to kernel functions. Therefore, reading or writing a `proc` file results in a function call invoked in kernel space, which can be used to read/write kernel parameters and therefore monitor and/or control the Linux networking stack. The Linux networking stack also provides many virtual files inside `/proc`, which can be used to monitor and/or control the networking stack.

This article is divided into three sections—monitoring, management/control and security, though sometimes it is difficult to make a clear distinction between these three aspects.

Monitoring

Here, we will cover some of the methods to monitor the TCP/IP networking stack.

Packet monitoring: The primary